Section 5.3 Geometric probability

几何概率

[AIME-2010-II-Q02] | Part A (Q1-Q5)

Choose a uniform point in a unit square, and let d be its distance to the nearest side. Reduce $\Pr(1/5 \le d \le 1/3) = m/n$ and find $m+n$.

在单位正方形内均匀选一点，d 为它到最近边的距离。若 $\Pr(1/5 \le d \le 1/3) = m/n$ 为最简分数，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.3 Geometric probability

几何概率

[AIME-2017-II-Q03] | Part A (Q1-Q5)

Choose a point uniformly by area inside triangle with vertices $(0,0),(12,0),(8,10)$. If its probability of being nearer $(12,0)$ than either other vertex is reduced p/q, find $p+q$.

在顶点为 $(0,0),(12,0),(8,10)$ 的三角形内按面积均匀取点。若该点到 $(12,0)$ 比到其余两个顶点都近的概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.3 Geometric probability

几何概率

[AIME-2020-II-Q02] | Part A (Q1-Q5)

Choose (x, y) uniformly from the unit square. Let $Q = (5/8, 3/8)$. The probability that the line from Q to (x, y) has slope at least $1/2$ is m/n reduced. Find $m + n$.

在单位正方形内均匀选点 (x, y)。固定 $Q = (5/8, 3/8)$，连接两点的直线斜率至少为 $1/2$ 的概率约分为 m/n，求 $m + n$。

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Section 5.3 Geometric probability

几何概率

[AIME-2004-I-Q10] | Part B (Q6-Q10)

A radius-one circle is placed with its center uniformly distributed among positions keeping it inside a 36×15 rectangle. The probability that it does not meet a given diagonal is m/n reduced. Find $m+n$.

半径一的圆完全位于 36×15 矩形内，圆心在全部允许位置中均匀分布。不与给定对角线相交的概率约分为 m/n，求 $m+n$。

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Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q06] | Part B (Q6-Q10)

An isosceles triangle with equal base angles x degrees is inscribed in a circle. Two independent uniformly sampled circumference points determine a chord. Its probability of meeting the triangle is $14/25$. Find the largest possible x minus the smallest.

一个等腰三角形内接于圆，两底角均为 x 度。在圆周上独立均匀取两点形成弦，该弦与三角形相交的概率为 $14/25$。求 x 的最大可能值与最小可能值之差。

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Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q08] | Part B (Q6-Q10)

Independently choose a, b uniformly from $(0, 75)$ degrees. Let $OP = 200$. On the same side of OP, choose Q, R with right angles OQP, ORP and angles $POQ = a, POR = b$. If $\Pr(QR \leq 100) = m/n$ in lowest terms, find $m + n$.

角度 a, b 独立均匀取自 $(0, 75)$ 度，$OP = 200$。在 OP 同侧取 Q, R，使 OQP, ORP 均为直角，且 $POQ = a, POR = b$。若 $\Pr(QR \leq 100) = m/n$ 为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.3 Geometric probability

几何概率

[AIME-2023-II-Q06] | Part B (Q6-Q10)

Choose two independent uniform points in $L = [0,1]^2 \cup ([0,1] \times [1,2]) \cup ([1,2] \times [0,1])$. The chance that their midpoint also belongs to L is the reduced fraction m/n. Find $m+n$.

在区域 $L = [0,1]^2 \cup ([0,1] \times [1,2]) \cup ([1,2] \times [0,1])$ 内独立均匀选取两点。两点中点仍在 L 内的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.3 Geometric probability

几何概率

[AIME-2003-I-Q11] | Part C (Q11-Q15)

Choose x uniformly in $(0^\circ, 90^\circ)$. Let p be the probability that $\sin^2 x, \cos^2 x, \sin x \cos x$ fail to form a nondegenerate triangle. If $p = d/n$ where d is the degree measure of $\arctan m$ and positive integers m, n satisfy $m + n < 1000$, find $m + n$.

在 $(0^\circ, 90^\circ)$ 均匀选角 x。设 $\sin^2 x, \cos^2 x, \sin x \cos x$ 不能构成非退化三角形的概率为 p。若 $p = d/n$，d 为 $\arctan m$ 的度数，正整数 m, n 满足 $m + n < 1000$，求其和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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